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EDUCATION

ETH Zürich

09/2025 – Present

Masters in Electrical Engineering & Information Technology

Zürich, Switzerland

- *Relevant Coursework:* Principles of Distributed Computing, System-on-Chip: Data Analytics & Machine Learning, VLSI I, VLSI II, Machine Learning for Microcontrollers, Probabilistic AI

Vellore Institute of Technology

08/2020 – 08/2024

Bachelor of Technology in Electronics and Communication Engineering

Chennai, India

EXPERIENCE

Student Research Assistant, Center for Project-Based Learning

02/2026 – 05/2026

Advisor: Prof. Michele Magno

ETH Zürich, Switzerland

- Built a real-time obstacle avoidance pipeline on an ALIF MCU, combining object segmentation and monocular depth estimation to hit sub-10 ms end-to-end latency.
- Optimized TFLite models and wrote bare-metal C firmware to run inference within a 1 mW power budget.

Software Engineer I, Visteon Corporation

08/2024 – 07/2025

Full-Time

Chennai, India

- Profiled and optimized C++ modules in Jaguar Land Rover's IVI real-time system, improving throughput by 78.3% without breaking existing timing guarantees, and shipped the change to production.
- Integrated an on-device AI inference pipeline on Jetson Nano to automate sensor-triggered script executions.

PROJECTS

NE16 Deeploy Integration and Benchmarking

05/2026 – 08/2026

- Benchmarked kernel and full-model performance on GAP9 through Deeploy, an ML compiler that compiles PyTorch models to bare-metal C for a hardware accelerator.
- Traced a throughput bottleneck to a redundant memory-layout transform repeated across nearly every layer and rewrote the compiler pass to fold it out, lifting throughput by 1.48×.
- Characterized where the NE16 datapath saturates across channel counts and feature-map sizes, showing throughput only picks up meaningfully from 16 channels onward.

FIR Filter Hardware Accelerator for Croc SoC

04/2026 – 06/2026

- Designed a parameterizable 32-tap FIR filter accelerator (1-16 MAC units) for the Croc SoC in SystemVerilog and connected it over the OBI interconnect, using a FIFO-decoupled writeback path to overlap compute with memory access.
- Compared cycle count, area, and power across 5 MAC configurations and identified 8 MACs as the Pareto-optimal design, a 90.5× speedup over software.
- Took the design through synthesis, place-and-route, and DRC/LVS closure at 80 MHz on the IHP SG13G2 process using Yosys and OpenROAD.
- Verified functional correctness against a Python golden model across signal lengths from 36 to 540 samples, including edge-case coefficients and signal values.

Universal Semantic World Models - Collaboration with Google

02/2026 – 06/2026

- Built a multi-view semantic fusion pipeline in Python that lifts DINOv3 features onto reconstructed 3D points and aggregates them with SLERP, enabling open-vocabulary 3D scene queries without retraining.
- Defined cosine dispersion to measure cross-view semantic flickering, then compared averaging, weighted SLERP, and mask-guided SLERP fusion using PyTorch and VGGT.
- Found mask-guided fusion improved segmentation mIoU by 3.5 points (51.5 to 55.0) over simple averaging on 50k Replica points, at the cost of lower accuracy on small object classes.

TECHNICAL SKILLS

Programming Languages & Tools: Python, C++, C, SQL, SystemVerilog, Git, Linux, Docker, CMake

Hardware: GAP9 (RISC-V), Jetson Nano, ALIF, MAX78000, Raspberry Pi, ESP-32, Arduino

Frameworks & Libraries: PyTorch, TensorFlow, TFLite, Keras, OpenCV

Language Proficiencies: English, Tamil (Native), German (Beginner), French (Beginner)